Experiments 9–10: completing a fact switches the heads back on

GPT-2 small · BOS + 32 tokens · **resting** = late-layer (6–11) attention parked on the first token (BOS) · maps show common filler · examples are real rows: BOS start gray random filler blue inserted words orange final token, where attention is measured

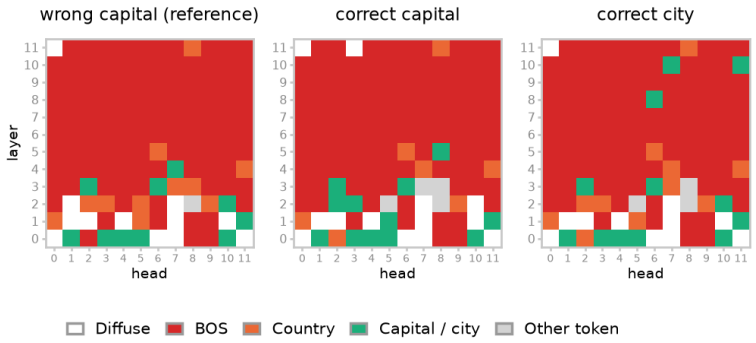
Experiment 9 · the capital is the final token

11 countries × 11 capitals · 200 rows per filler type

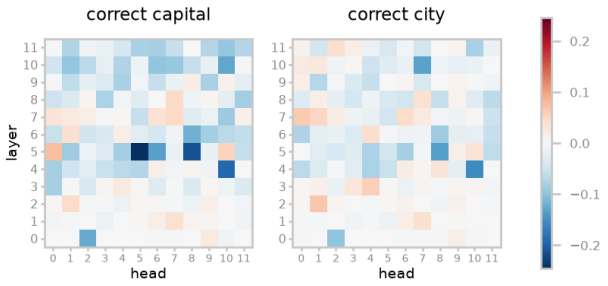
correct BOS she sc sa … 27 random tokens … France Paris

wrong BOS she sc sa … 27 random tokens … Japan Paris

What each head attends to most



Change in BOS attention (red = more resting)



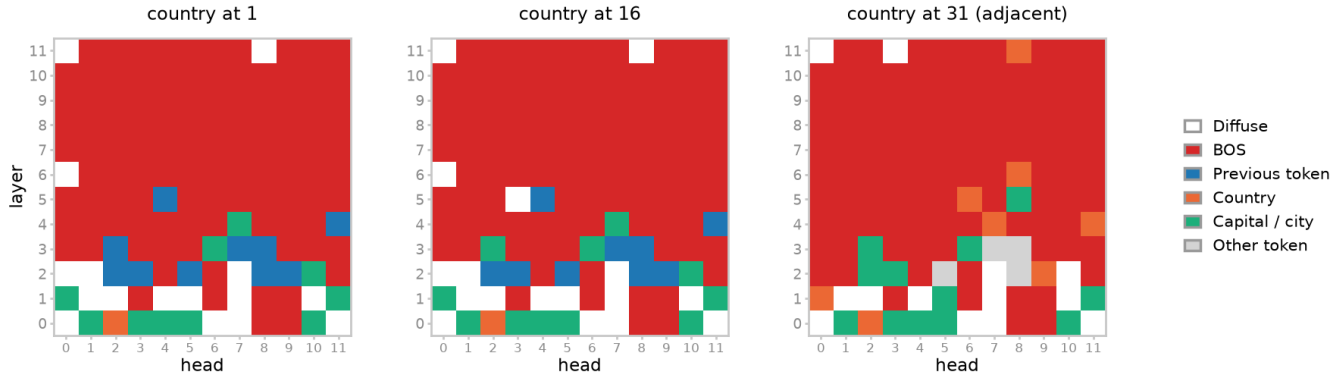
Experiment 10 · the country anywhere earlier

496 placements · 50 rows per filler type

correct BOS France sc sa … dis Paris | BOS she … ors France ign … dis Paris | BOS she sc … age France Paris

wrong BOS Japan sc sa … dis Paris | BOS she … ors Japan ign … dis Paris | BOS she sc … age Japan Paris

What each head attends to most (correct pairs)



Change in BOS attention, correct − wrong (blue = less resting)

